points (difference in boiling points is less than 25 K) fractional distillation is employed.

Fractional distillation is used in petrochemical industry to obtain different fractions of petroleum, to separate the different gases from air, to distill alcohols etc.

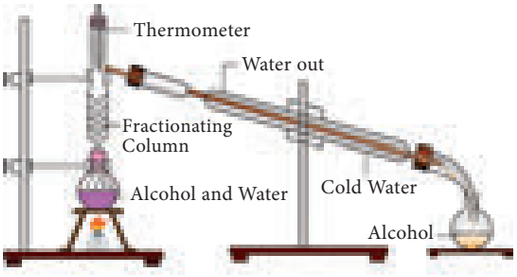


Figure 10.9 Fractional distillation

10.3.6 Chromatography

Before we discuss the technique we will take a look at the difference between the two important terms: **Absorption** and **Adsorption**

Adsorption is the process in which the particles of a substance is concentrated only at the surface of another substance.

Absorption is the process in which the substance is uniformly distributed throughout the bulk of another substance.

For example, when a chalk stick is dipped in ink, the surface retains the colour of the ink due to adsorption of coloured molecules while the solvent of the ink goes deeper into the stick due to absorption. Hence, on breaking the chalk stick, it is found to be white from inside.

Chromatography is also a separation technique. It is used to separate different components of a mixture based on their different solubilities in the same solvent. There are several types of chromatography based on the above basic principles. The simplest type is paper chromatography.

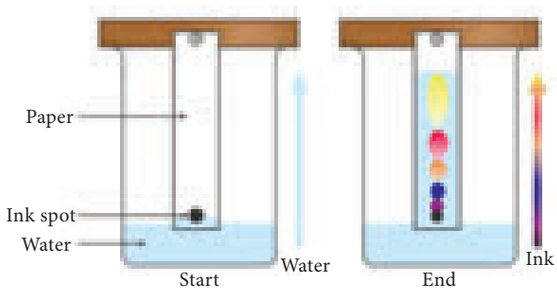


Figure 10.10 Paper chromatography

Paper chromatography

This method is used to separate the different coloured dyes in a sample of ink. A spot of the ink (e.g. black ink) is put on to a piece of chromatography paper. This paper is then set in a suitable solvent as shown in figure 10.10. The black ink separates into its constituent dyes. As the solvent moves up the paper, the dyes are carried with it and begin to separate. They separate because they have different solubility in the solvent and are adsorbed to different extents by the chromatography paper. The chromatogram shows that the black ink contains three dyes.

10.4 Solutions

A solution is a homogeneous mixture of two or more substances. In a solution, the component present in lesser amount by weight is called solute and the component present in larger amount by weight is called solvent.

In short, a solution can be represented as follows: solute + solvent ⟶ solution

Example: salt + water ⟶ salt solution

10.4.1 Types of solution

Based on the particle size of the solute, the solutions are divided into three types. Let us study them through an activity.

Activity 5

Take bottles containing sugar, starch and wheat flour. Add one tea spoon full of each one to a glass of water and stir well. Leave it aside for about ten minutes. What do you observe?



IX_SCI_EM_Unit-10_CHE.indd 119 06-02-2020 11:47:34